

Chap. 8 Central Processing Unit

• 8-1 Introduction

– 3 major parts of CPU : *Fig. 8-1*

- 1) Register Set
- 2) ALU
- 3) Control

– Design Examples of simple CPU

- Hardwired Control : *Chap. 5*
- Microprogrammed Control : *Chap. 7*

– In this chapter : *Chap. 8*

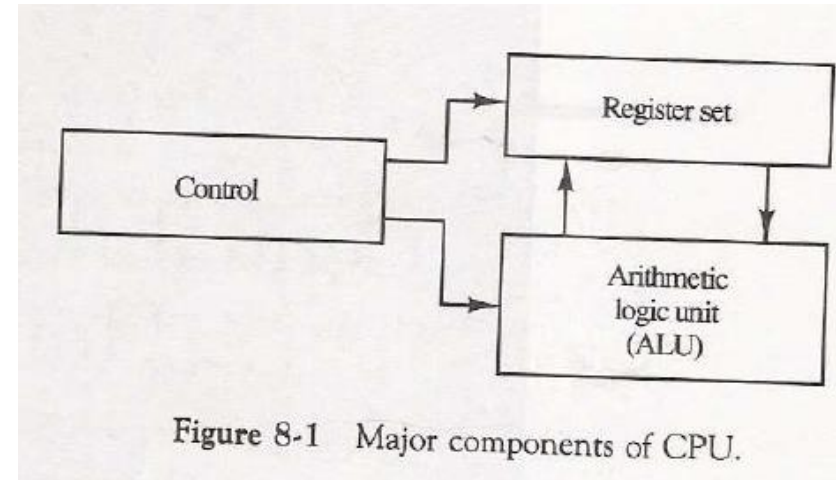
Computer Architecture as seen by the programmer

- Describe the organization and architecture of the CPU with *an emphasis on the user's view of the computer*
- User who programs the computer in machine/assembly language *must be aware* of

- 1) Instruction Formats
- 2) Addressing Modes
- 3) Register Sets

Chap. 8 CPU

- The last section presents the concept of *Reduced Instruction Set Computer (RISC)*



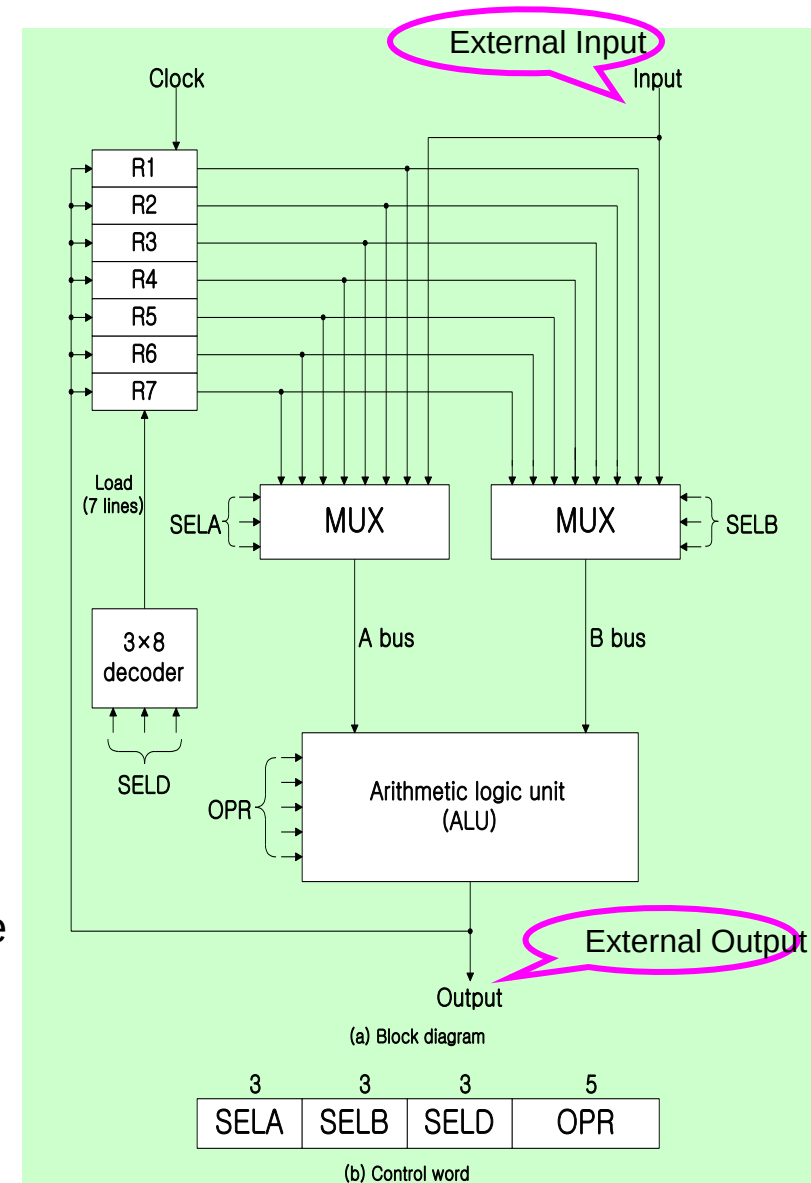
- Architecture Types (based on Instruction set).
 - RISC
 - CISC

- Architecture Types (based on memory usage)
 - Nonembedded Architecture (Von Neuman)
 - Embedded Architecture (Harvard)



8.2 General Register Organization

- CPU Registers
 - Memory locations are needed for storing *pointers, counters, return address, temporary results, and partial products* during multiplication (*in the programming examples of Chap. 6*)
 - Memory access is the most time-consuming operation in a computer
 - More convenient and efficient way is to store intermediate values in processor registers
- Bus organization for 7 CPU registers : **Fig. 8-2**
 - **2 MUX** : select one of 7 register or external data input by **SELA** and **SELB**
 - **BUS A and BUS B** : form the inputs to a common ALU
 - **ALU** : **OPR** determine the arithmetic or logic microoperation
 - The result of the microoperation is available for external data output and also goes into the inputs of all the registers
 - **3 X 8 Decoder** : select the register (by **SELD**) that receives the information from ALU



$$R1 \leftarrow R2 + R3$$

- Binary selector input :
 - 1) MUX A selector (**SELA**) : to place the content of R2 into BUS A
 - 2) MUX B selector (**SELB**) : to place the content of R3 into BUS B
 - 3) ALU operation selector (**OPR**) : to provide the arithmetic addition $R2 + R3$
 - 4) Decoder selector (**SELD**) : to transfer the content of the output bus into R1
- Control Word
 - 14 bit control word (*4 fields*) : **Fig. 8-2(b)**
 - **SELA** (*3 bits*) : select a source register for the A input of the ALU
 - **SELB** (*3 bits*) : select a source register for the B input of the ALU
 - **SELD** (*3 bits*) : select a destination register using the 3 X 8 decoder
 - **OPR** (*5 bits*) : select one of the operations in the ALU
 - Encoding of Register Selection Fields : **Tab. 8-1**  **Tab. 8-2**
 - **SELA** or **SELB** = **000 (Input)** : MUX selects the external input data
 - **SELD** = **000 (None)** : no destination register is selected but the contents of the output bus are available in the external output
 - Encoding of ALU Operation (**OPR**) : **Tab. 8-2**
- Examples of Microoperations : **Tab. 8-3**
 - TSFA (Transfer A) : $R7 \leftarrow R1, \text{External Output} \leftarrow R2, \text{External Output} \leftarrow \text{External Input}$
 - XOR : $R5 \leftarrow 0 (XOR R5 \oplus R5)$

} **Tab. 8-1**

Tab. 8-2

Control Word - Control Memory -
Microprogrammed Control



TABLE 8-1 Encoding of Register Selection Fields

Binary Code	SELA	SELB	SELD
000	Input	Input	None
001	R1	R1	R1
010	R2	R2	R2
011	R3	R3	R3
100	R4	R4	R4
101	R5	R5	R5
110	R6	R6	R6
111	R7	R7	R7

TABLE 8-2 Encoding of ALU Operations

OPR Select	Operation	Symbol
00000	Transfer A	TSFA
00001	Increment A	INCA
00010	Add $A + B$	ADD
00101	Subtract $A - B$	SUB
00110	Decrement A	DECA
01000	AND A and B	AND
01010	OR A and B	OR
01100	XOR A and B	XOR
01110	Complement A	COMA
10000	Shift right A	SHRA
11000	Shift left A	SHLA

